



LARAVEL CHEAT SHEET

Core setup, initialization, operations & integration quick reference

PART 1: SETUP & CORE

01. SETUP & ARTISAN

Installation

```
composer create-project laravel/laravel myapp
php artisan serve
php artisan make:controller UserController --...
php artisan make:model User -m (with migrati...
php artisan migrate
php artisan tinker (REPL)
```

04. ELOQUENT ORM

Workflow

```
User::all();
User::find(1);
User::where('active', true)->orderBy('name')-...
User::firstOrCreate(['email' => $email], ['na...
$user->save(); // $user->delete();
User::create(['name' => 'Alice']); // mass a...
User::chunk(100, fn($users) => ...); // batc...
```

02. ROUTING

Architecture

```
Route::get('/users', [UserController::class, ...
Route::post('/users', [UserController::class, ...
Route::apiResource('users', UserController::c...
Route::get('/profile', fn() => view('profile'...
Route::prefix('api')->group(fn() => { Route::...
```

05. MIGRATIONS

CRUD API

```
Schema::create('users', function (Blueprint $...
$table->id();
$table->string('name');
$table->string('email')->unique();
$table->timestamps();
});
php artisan migrate / migrate:rollback / ...
```

03. CONTROLLERS

YAML / Config

```
class UserController extends Controller {
    public function index() {
        return User::paginate(15);
    }
    public function store(StoreUserRequest $req) {
        $user = User::create($req->validated());
        return response()->json($user, 201);
    }
}
```

06. BLADE TEMPLATES

Integration

```
{{ $user->name }} {!! $html !!}
@if($user->isAdmin()) Admin @endif
@foreach($users as $user) {{ $user->name }} @...
@extends('layouts.app')
@section('content') ... @endsection
@component('alert', ['type' => 'error'])
```



SECURITY, MONITORING & OPERATIONS

Production hardening, observability, performance tuning & CLI reference

PART 2: OPS & SECURITY

07. MIDDLEWARE

Hardening

```
php artisan make:middleware EnsureUserIsAdmin
public function handle(Request $req, Closure ...
if (!$req->user()->isAdmin()) abort(403);
return $next($req);
}
// Register in app/Http/Kernel.php or bootstr...
Route::middleware(['auth', 'admin'])->group(...
```

10. QUERY BUILDER

Unit Tests

```
DB::table('users')
->select('name', 'email')
->where('active', true)
->orderBy('name')
->paginate(15);
DB::table('users')->insert(['name'=>'Alice']);
DB::transaction(function() { ... });
```

08. VALIDATION

Observability

```
$validated = $request->validate([
'name' => 'required|string|max:255',
'email' => 'required|email|unique:users',
'age' => 'nullable|integer|min:0',
]);
// Form Request:
class StoreUserRequest extends FormRequest {
public function rules(): array { ... }
}
```

11. COMMON ARTISAN COMMANDS

Commands

```
php artisan make:model/controller/middleware/...
php artisan route:list / cache:clear / co...
php artisan queue:work / queue:listen
php artisan storage:link / key:generate
php artisan schedule:work (runs scheduler lo...
```

09. ELOQUENT RELATIONSHIPS

Optimization

```
// User.php:
public function orders(): HasMany {
return $this->hasMany(Order::class);
}
public function role(): BelongsTo {
return $this->belongsTo(Role::class);
}
// Eager loading:
User::with('orders', 'role')->get();
```

12. COMMON GOTCHAS

Warning

Mass assignment: \$fillable or \$guarded on model
N+1 queries: always use with() for relationships
APP_KEY required generate with artisan key:generate
Storage link needed for public disk file access
Eloquent soft deletes need SoftDeletes trait + deleted_at column